Linguagens Script

<Apresentação>

Departamento de Engenharia Informática e de Sistemas

Cristiana Areias < cris@isec.pt >

2022/2023

Docentes

- Teóricas
  - Cristiana Areias < cris@isec.pt > – ***Docente Responsável***
- Práticas
  - Cristiana Areias
  - Daniel Moreira< daniel.moreira@isec.pt >
  - Pedro Reis <preis@isec.pt >
  - Pedro Sá <pedro.ss@isec.pt >
  - Tiago Figueira <tiago@isec.pt >

Informações da UC

- ***1º** Ano – **2º** Semestre*
- *Créditos ECTS: 4*
- *Modo de Ensino*
  - Presencial
- *Horas totais de contacto:*
  - Componente teórico: 15 horas
  - Componente prática-laboratorial: 30 horas
- *Horas de estudo não acompanhado:*
  - Estudo: 22 horas
  - Trabalhos/Trabalhos de Grupo:40h

Objetivos e Competências

- Aprendizagem dos **fundamentos das linguagens script;**
- Aprendizagem da **linguagem *JavaScript*** no contexto do *front end development;*
- Apresentação de bibliotecas e *frameworks* fortemente utilizadas na indústria para construção de aplicações web dinâmicas, com foco especial da biblioteca ***React JS***;
- **Aplicação prática** dos conhecimentos abordados nas aulas teóricas;
- **Aprendizagem autónoma** e procura de soluções novas;
- **Raciocínio crítico**, através da construção de uma aplicação web dinâmica.

Programa Genérico

- Componente Teórica
  - Visão geral das linguagens script
    - Diferenças com outras linguagens
    - Linguagens script em aplicações web
      - *Client-Side Scripting*
      - Bibliotecas e *frameworks*
  - *Linguagem Javascript*
  - *Biblioteca ReactJS*
  - *Linguagens scripts: outros tópicos*
    - Preocupações de segurança em *Client-Side*
    - Introdução a linguagens de script para *Server-Side*
- Componente Prática: Resolução de exercícios sobre *Javascript e ReactJS*

Metodologia de Avaliação

- A avaliação será realizada pelas seguintes componentes:
  - **Exame (12 Valores)**
  - **Trabalho Prático (8 Valores)**
  - **Classificação Final** = **0,6 * Exame + 0,4 * TP**
  - **Bónus de 1 valor**, a adicionar à nota final, a quem assistir ao *webminar* completo de "*Web Components - Write once, run anywhere*", que se a realizará durante a tarde do dia **17/05/2023** *(ver condições nas inscrições)*.
- **Exame**
  - Sem consulta
  - É obrigatória a obtenção de uma classificação igual ou superior a 35% para obter aprovação à disciplina
  - Conteúdos lecionados na componente teórica e prática

Metodologia de Avaliação

- **Trabalho Prático**
  - A implementar com base em ***React*.**
  - Grupos de 2 ou 3 alunos;
  - O **trabalho é obrigatório**, deste modo a não realização do trabalho implica a reprovação à disciplina.
  - Não existem mínimos no trabalho prático.
  - Não é possivel manter nota do trabalho prático realizado em anos anteriores;
- **Melhoria de Classificação**
  - Qualquer processo de melhoria de classificação da prova escrita deve verificar integralmente os requisitos definidos no regulamento de avaliação em vigor no ISEC/IPC.

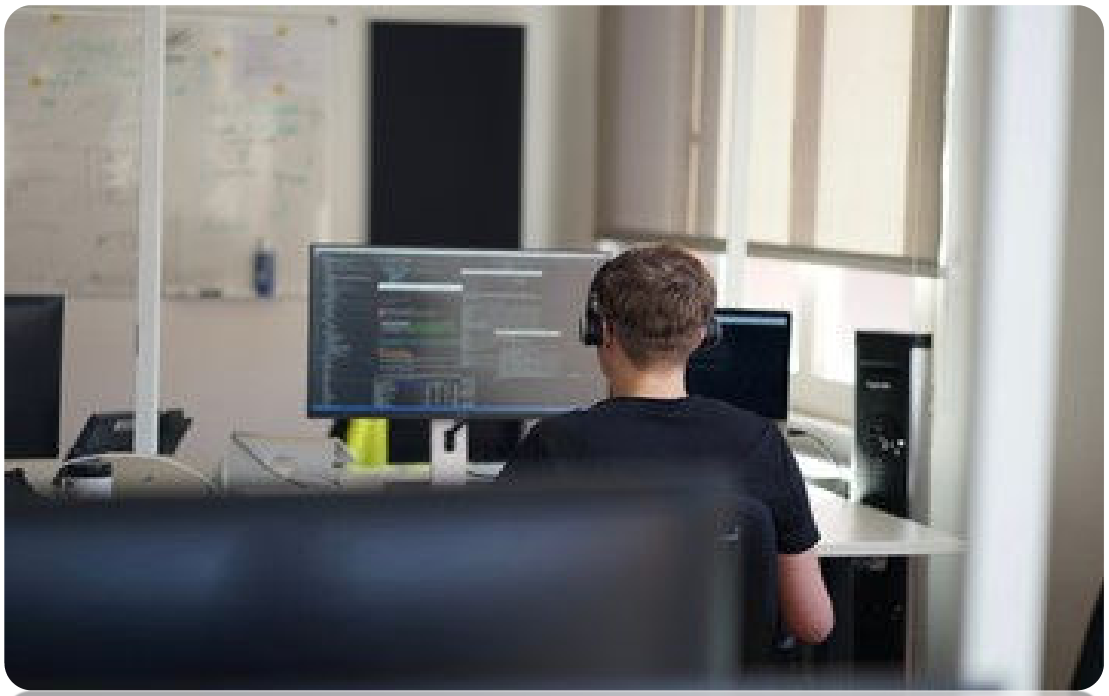
Bibliografia

- Material de apoio às aulas a disponibilizar no Inforestudante
- Internet é uma excelente fonte, e o Google um bom ponto de partida:
  - Javascript | MDN Web Docs (2022). JavaScript Guide. https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide
  - ReactJS (2023). ReactJ - The library for web and native user interfaces https://react.dev/
  - https://stackoverflow.com/
- Livros:
  - Flanagan, D. (2020) JavaScript: The Definitive Guide, 7th Edition, ISBN: 9781491952023, O'Reilly Media, Inc. (ISEC: 1ª-12-177)
  - Wieruch, R., (2018) The Road to learn React: Your journey to master plain yet pragmatic React.js, ISBN-10: 9781720043997, Independently published (ISEC1A-12-197)

Bibliografia

- Livros (cont.):
  - Chinnathambi, K. (2018). Learning React: A Hands-On Guide to Building Web Applications Using React and Redux, Second edition, ISBN: 9780134843582, Addison-Wesley Professiona (ISEC:1A-12-202).
  - Gordon, Z. (2020). *Javascript explained : step-by-step guide to the most common and reliable JS techniques,* ISBN 9798623901415, Houston, OSTraining. (ISEC: 1A -12-172)
  - Gordon, Z. (2018). *React explained,* ISBN 979-8-623901-41-5, Houston, OSTraining. (ISEC: 1A -12-203)
  - Robin N. (2018). *Learning PHP, MySQL & JavaScript : with jQuery, CSS & HTML5*, ISBN: 9781491978917, Beijing, O'Reilly. (ISEC: 1A-12-201)

Linguagens Script



Algum profissional?

Porquê JavaScript?





Porquê JavaScript?



https://i.redd.it/1dqd5hfq5sd01.jpg

Porquê JavaScript?

- Das mais populares e mais utilizadas linguagens de programação atualmente.
  - **JavaScript é usado por 97.9% de todos os sites web.**

https://w3techs.com/technologies/cross/client_side_language/javascript_library

According to *Stack Overflow's 2020 Developer Survey*, **JavaScript** currently stands as the most commonly-used language in the world **(69.7%),** followed by HTML/CSS (62.4%), SQL (56.9%), Python (41.6%) and Java (38.4%).

https://bootcamp.berkeley.edu/blog/most-in-demand-programming-languages/

JavaScript is the queen of programming languages

JavaScript is the most popular programming language by some distance, with nearly 14M developers using it globally. More importantly, the JavaScript community has been growing in size consistently for the past three years. Between Q4 2017 and Q1 2021, more than 4.5M developers joined the community - the highest growth in absolute terms across all languages. Even in software sectors where JavaScript is not among developers' top choices, like data science or embedded development, about a fourth of developers use it in their projects.



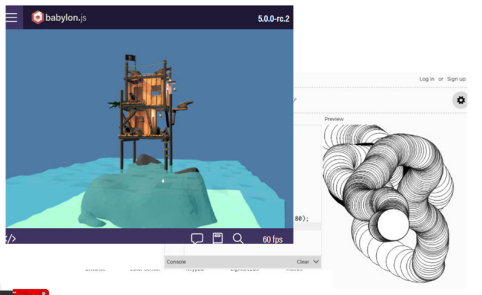
https://slashdata-website-cms.s3.amazonaws.com/sample_reports/ll65ncm-kfuTOtls.pdf

- Não só na web, mas utilizada em vários domínios!

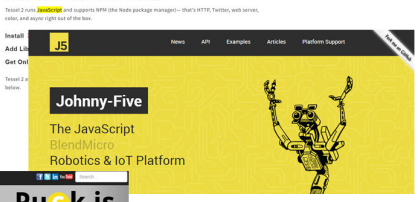
Linguagens Script

Porquê JavaScript?

- Browser
  - DOM (*DOM API, zepto.js, jQuery.js,…*)
  - Canvas & WebGL (*Canvas API, p5.js, three.js, Babylon.js,…*)
  - WebAudio (*WebAudio API, Tone.js, Tonal.js, Babylon.js,…*)
  - MediaCapture
  - …
- Hardware
  - Tessel (https://tessel.io)
    - Executa Node.js em sistemas baseados em linux
  - Espruino (https://espruino.com)
    - Interpretador Javascript Embebido
  - Johnny-Five (http://johnny-five.io/
    - Executa em varios tipos de hardware
  - WebMIDI (https://webmidijs.org/)
    - Acessivel no browser sem qualquer *plugin*







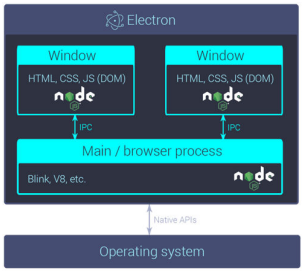




- AI e *Machine Learning*
  - Bibliotecas Client-side (*ml5.js, tensoftflow.js*, pico.ks, tracking.js,…)
  - Serviços Web (*Google Cloud Vision, Amazon Rekognition AWS, Azure Cognitive Services*)
  - Browser Buit-in (*Web Speech API*)

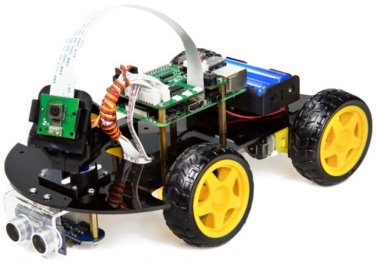
Porquê JavaScript?

- **Aplicações:**
  - Desenvolvimento web e aplicações Web
  - Jogos
  - Aplicações desktop nativas (Electron)
  - Aplicações Móveis (*React Native,* ionic)
  - Servidores web (Ex. *Node.js*)
  - Aplicações para *Smartwatch (Ex. Pebble.js*)
  - Arte (https://javascriptforartists.com)
  - Programação de Drones, *quadcopters, robots* (Com recurso ao *Node.js*).





Raspberry PI Robot Car Controlled by Node.js



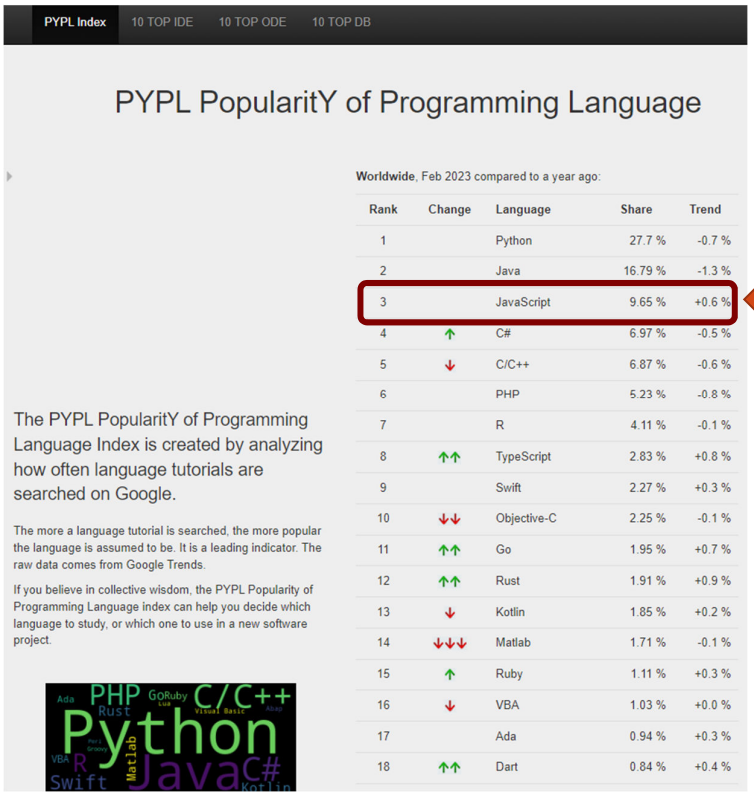
https://medium.com/swlh/raspberry-pi-robot-car-controlled-by-node-js-3d9c27c5b16f



Linguagens Script

Porquê JavaScript?

The more a language tutorial is searched**, the **more popular the language is assumed to be**. It is a leading indicator. The raw data comes from **Google Trends.



https://pypl.github.io/PYPL.html

Porquê JavaScript?

TIOBE Programming Community Index Definition

There are 25 search engines that are used to calculate the TIOBE index.

TIOBE

Schedule a demo

Feb 2023	Feb 2022	Change	Programming Language	Ratings	Change
1	1		Python	15.49%	+0.16%
2	2		C	15.39%	+1.31%
3	4	↑	C++	13.94%	+5.93%
4	3	↓	Java	13.21%	+1.07%
5	5		C#	6.38%	+1.01%
6	6		Visual Basic	4.14%	-1.09%
7	**7**		**JavaScript**	**2.52%**	**+0.70%**
8	10	↑	SQL	2.12%	+0.58%
9	9		Assembly language	1.38%	-0.21%
10	8	↓	PHP	1.29%	-0.49%
11	11		Go	1.11%	-0.12%
12	13	↑	R	1.08%	-0.04%
13	14	↑	MATLAB	0.99%	-0.04%

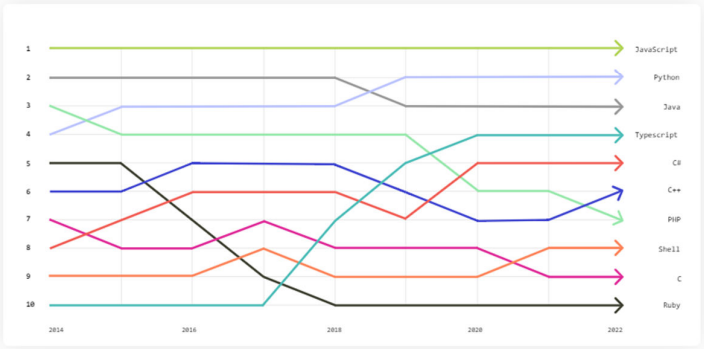
https://www.tiobe.com/tiobe-index/

Porquê JavaScript?

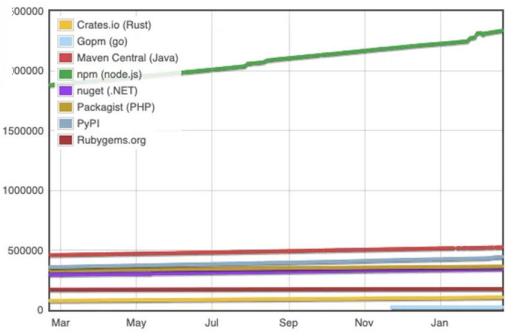
Top languages used in 2022

JavaScript continues to reign supreme and Python held steady in the second place position over the past year in large part due to its versatility in everything from development to education to machine learning and data science.

TypeScript also held firm in fourth place year-over-year. Notably, PHP dropped from sixth to seventh place in 2022.



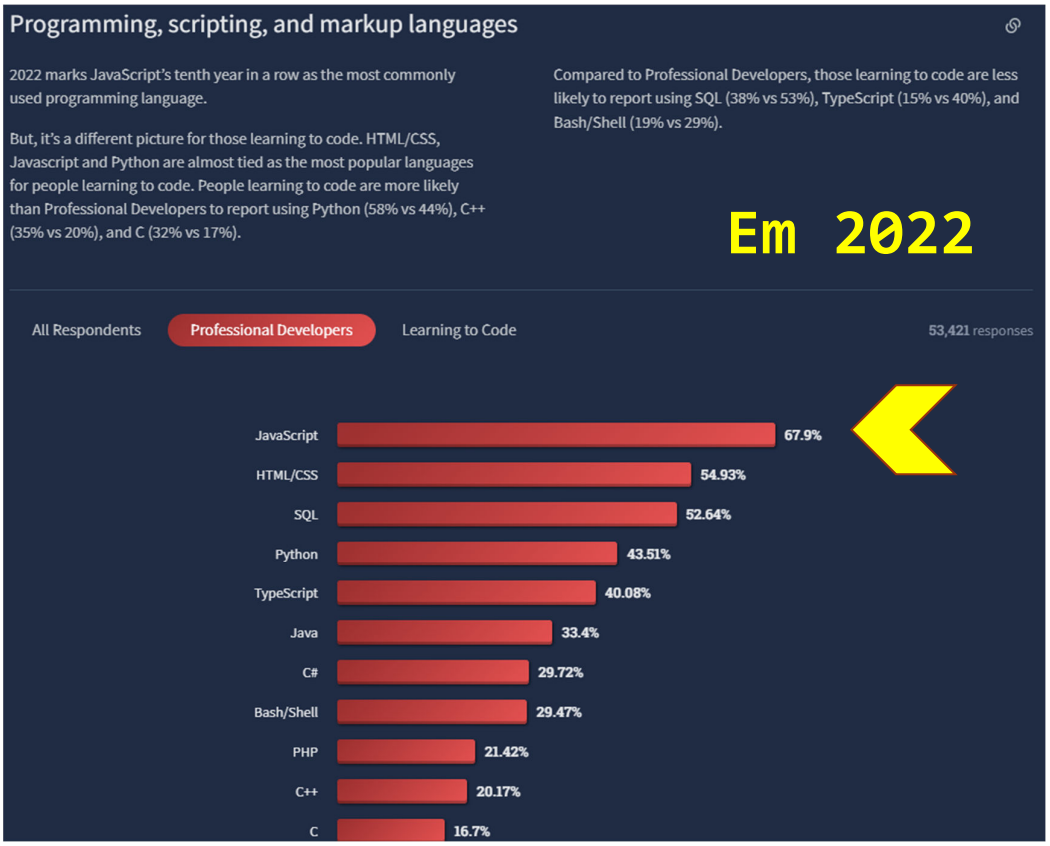
GitHub Language Popularity Rankings. Source: GitHub



Module Counts by Package Repository (NPM = JavaScript). Source: Module Counts

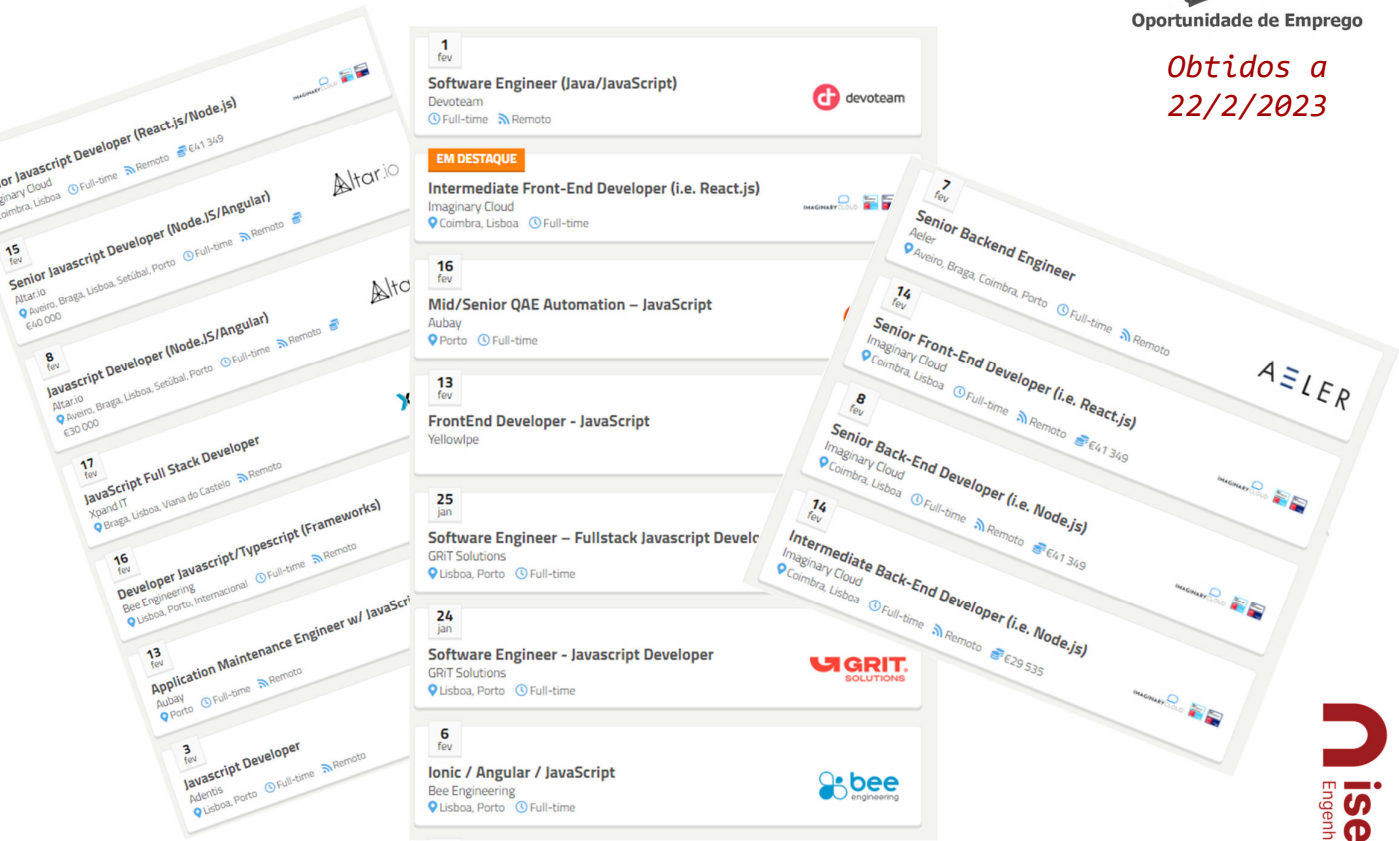
http://www.modulecounts.com/

Porquê JavaScript?



https://insights.stackoverflow.com/survey/

Javascript: Ofertas



https://www.itjobs.pt/

Bibliotecas e Frameworks

- Existem várias frameworks e bibliotecas (front-end e/ou back-end) que permitem a implementação rápida e fácil de aplicações modernas em grande escala
- Muitas baseadas **100% em Javascript**
- **No contexto da UC de Linguagens Script**
  - Bibliotecas UI
  - Frameworks Front-End















- Porque aprender JavaScript e não apenas a biblioteca/framework?

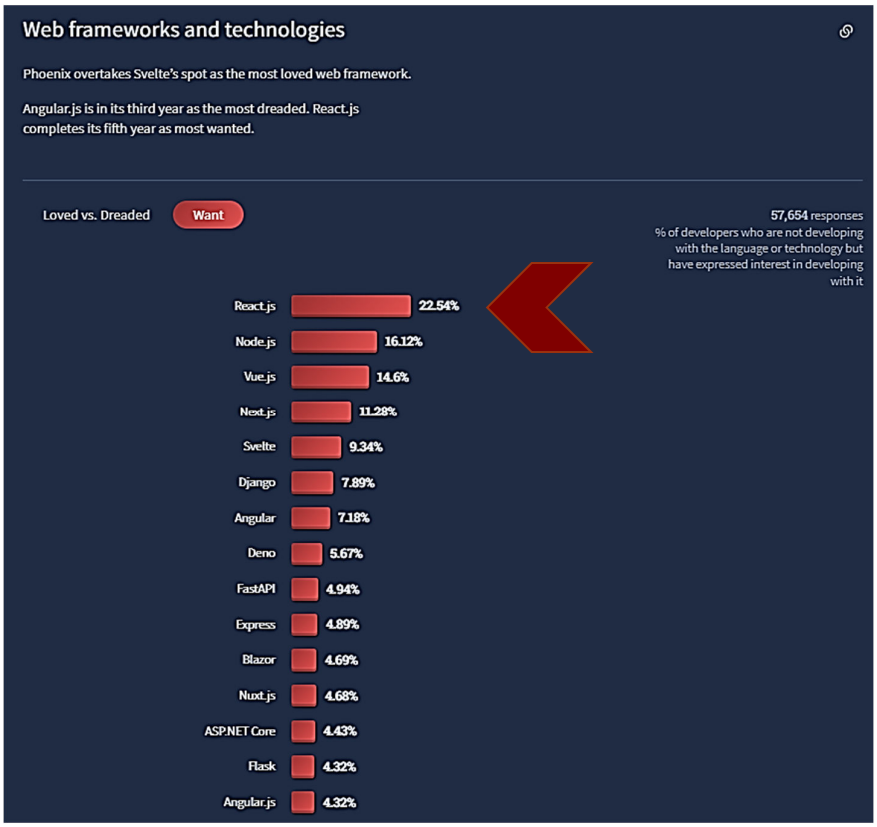
Frameworks Front-End/Bibliotecas UI

	Angular	React	Vue	Svelte	Next.js
Data de Lançamento	**2016** Open Source MIT Licence	**2013** Open Source MIT Licence	**2014** Open Source MIT Licence	**2016** Open Source MIT Licence	**2016** Open Source MIT Licence
Site Oficial	angular.io	reactjs.org	vuejs.org	svelte.dev	nextjs.org
	Google	Facebook	Evan You	Rich Harris	Vercel
Usado por	**Google**, Wix	**Facebook**, Uber, Netflix	Alibaba, GitLab, Zoom	nyTimes, decathlon, chess.com	GitHub, Netflix, Binance
					 *Framework para React*

Porquê ReactJS?



Tendências...

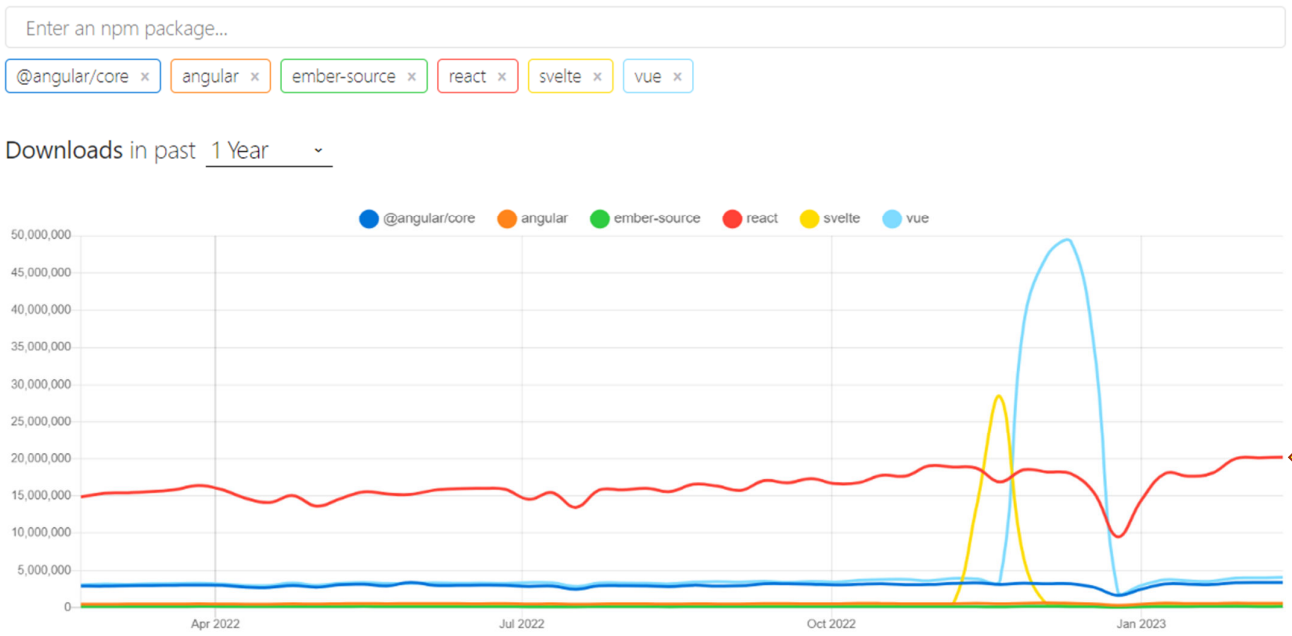


Em 2022

https://survey.stackoverflow.co/2022/

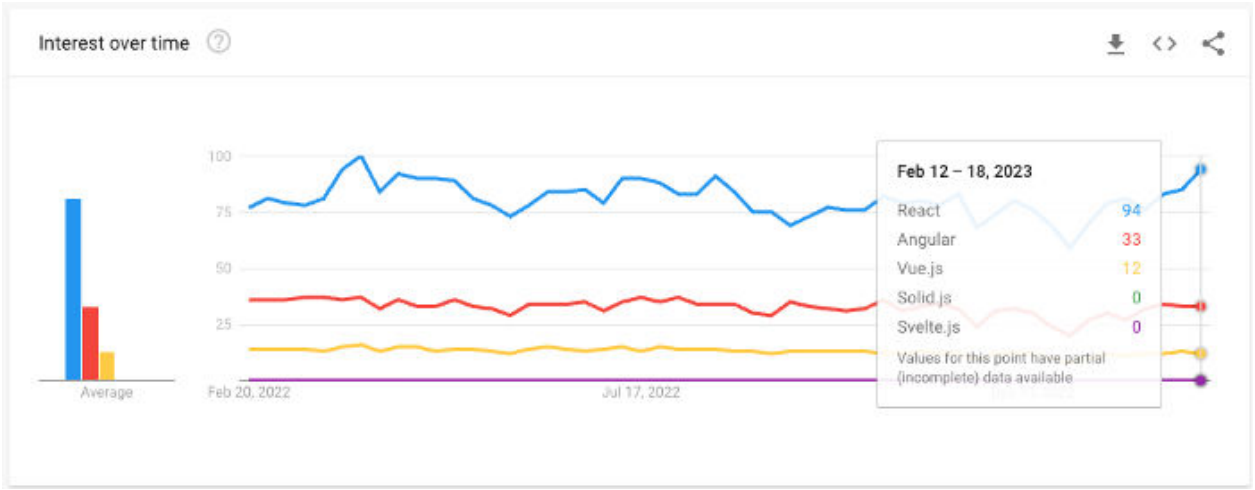
Tendências...

JavaScript package manager.



https://www.npmtrends.com/@angular/core-vs-angular-vs-ember-source-vs-react-vs-vue-vs-svelte

Tendências...

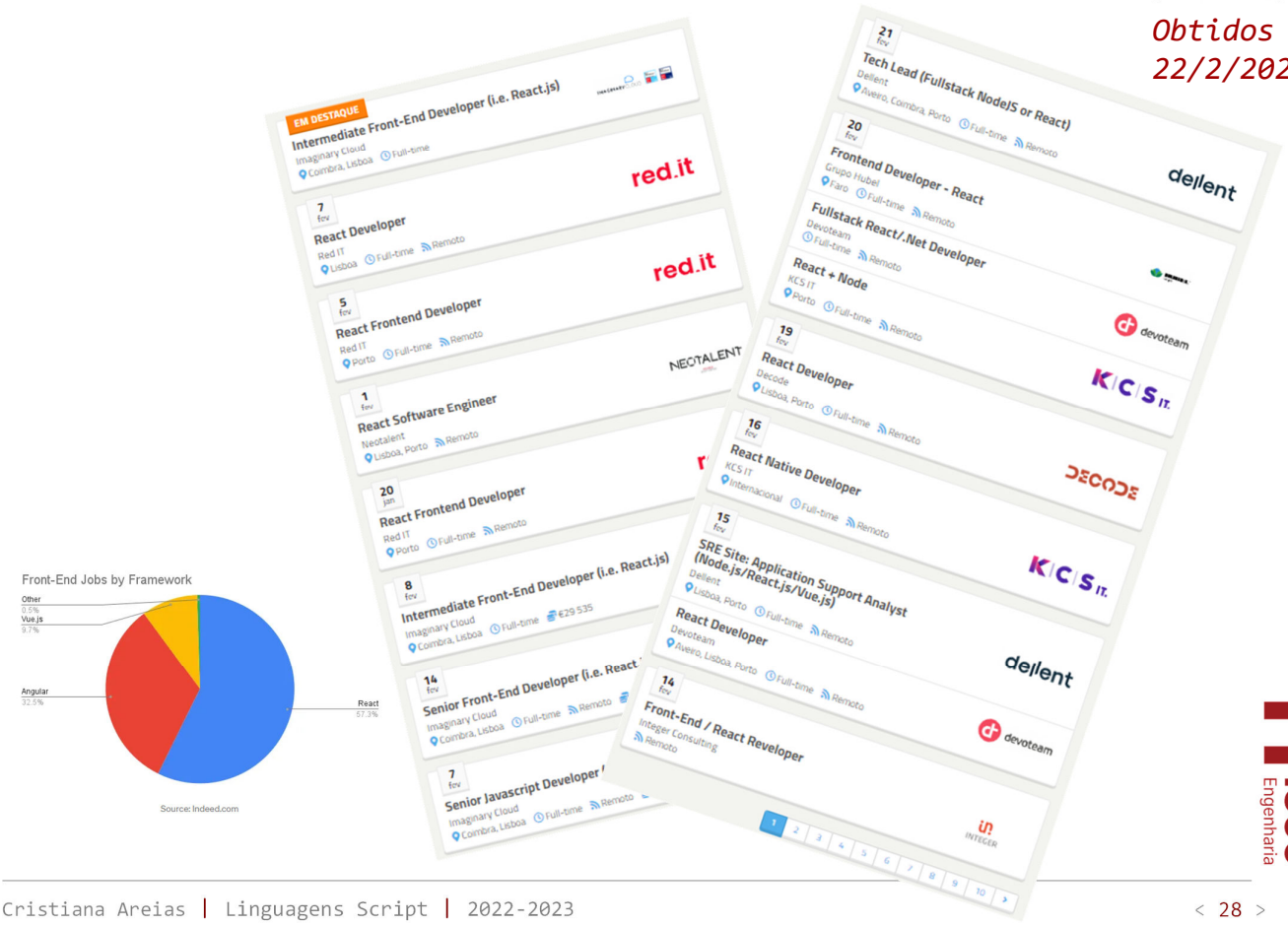


https://trends.google.pt/trends

ReactJS: Ofertas

Oportunidade de Emprego

Obtidos a 22/2/2023



JavaScript / React : Ofertas

Oportunidade de Emprego

Linguagens Script

2,538 Javascript React Jobs

Full Stack Developer Python JavaScript React
Client Server Ltd. Bracknell, UK
21 Feb
You'll be working across the full technology stack with a focus on Python backend work but with exposure across the front end including JavaScript and...

Lead Software Developer - C#, JavaScript, React, HTML
Gardiner Bros and Company (leathers) Ltd Hardwicke, UK
21 Feb
Lead Software Dev eloper - C#, JavaScript, React, HTML Salary £60,000 - £65,000 - Negotiable, depending on experience Hours Monday Friday 08:30 17:00...

Senior JavaScript/React Developer
WeDo London, UK
31 Jan
Senior JavaScript/React Developer | Senior Developer London // Fully Remote Up to £100,000 + Benefits I am lucky enough to be partnered with one of the...

Front End Developer- JavaScript/ React- £30K-£50K- Hybrid
Oscar Associates (UK) Limited Derbyshire, UK
31 Jan
Front End Developer- JavaScript/ React- £30,000-£50,000- Hybrid- Derby A majorly exciting role for has become available for one skilled mid-level...

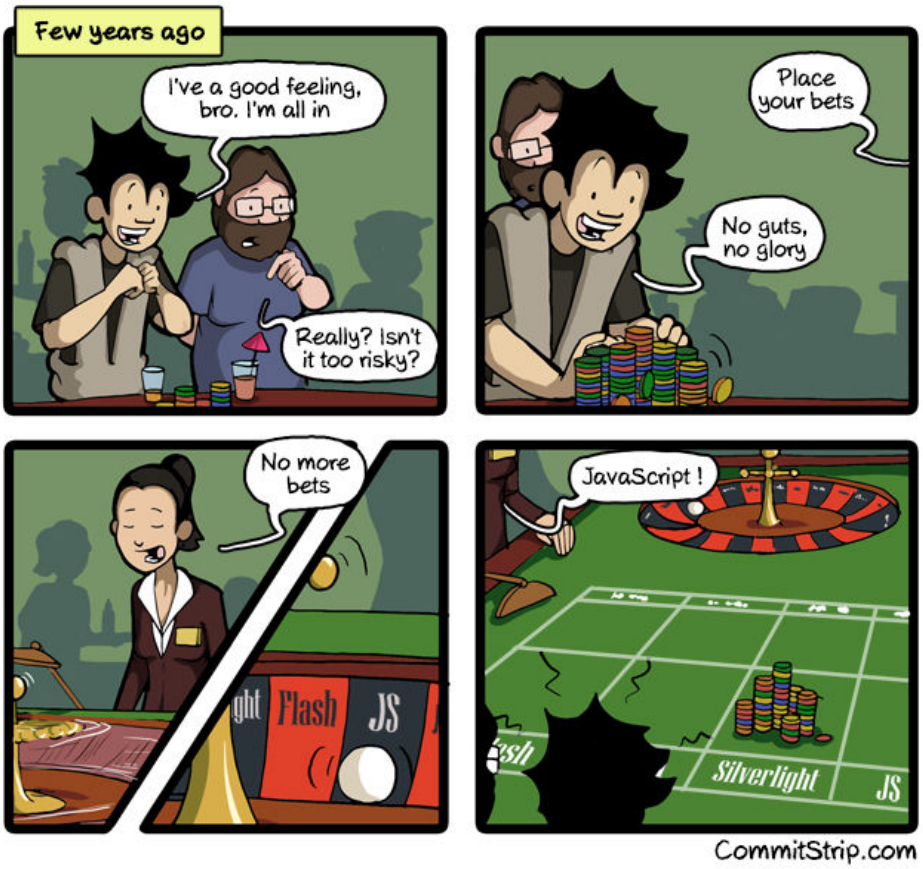
Web Developer JavaScript React Svelte
Client Server Ltd. London, UK
25 Jan
You have strong Vanilla JavaScript skills * You're happy to maintain and transition Angular.js code to React (you can pick React and Svelte up on the...

JavaScript Developer - React Svelte Outside IR35
SR2 London, UK
21 Feb
JavaScript / React / Svelte / Outside IR35 / Fully Remote Contract: Outside IR35 Length: 6 Months (Extensions likely) Rate: £500-£525 per day Location:...

Full Stack Developer- Node/JavaScript/React
Explore Group London, UK
31 Jan
React * NodeJS * Rest API If you feel like this will be a good fit for you, send me over a copy of your CV to

Senior software engineer (JavaScript, React, Front-end) - Construction tech
Trunk Manchester, UK
31 Jan
React powers our web app, for administration and reporting * Flutter drives our mobile apps and status boards. * Supabase is the open-source alternative...

https://www.ziprecruiter.co.uk/?utm_source=ziprecruiter-com-redirect



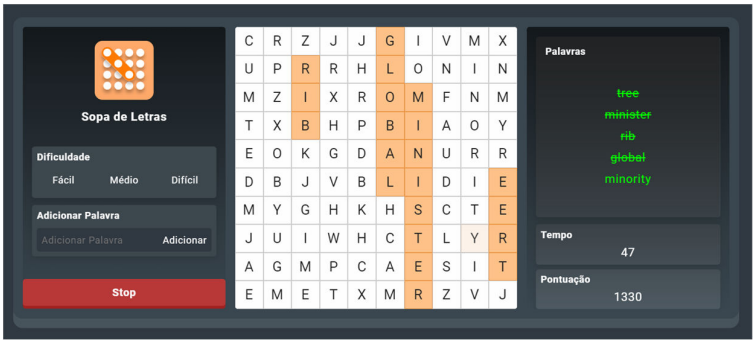
Linguagens Script

Jogo "Sopa de Letras"

› Trabalhos Práticos de Alunos - 2021/2022

"Sopa de Letras"



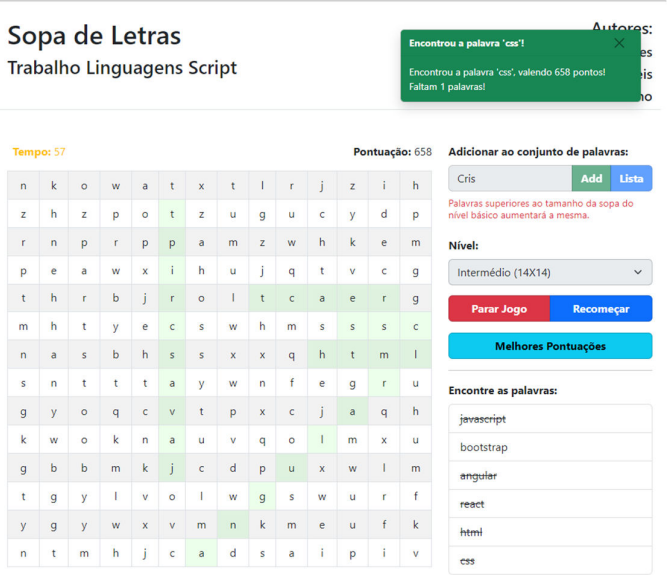


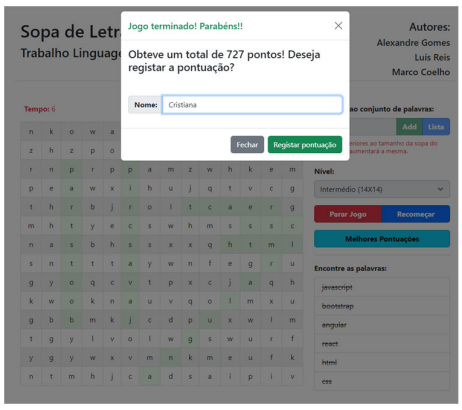


"Sopa de Letras"

Trabalhos Práticos

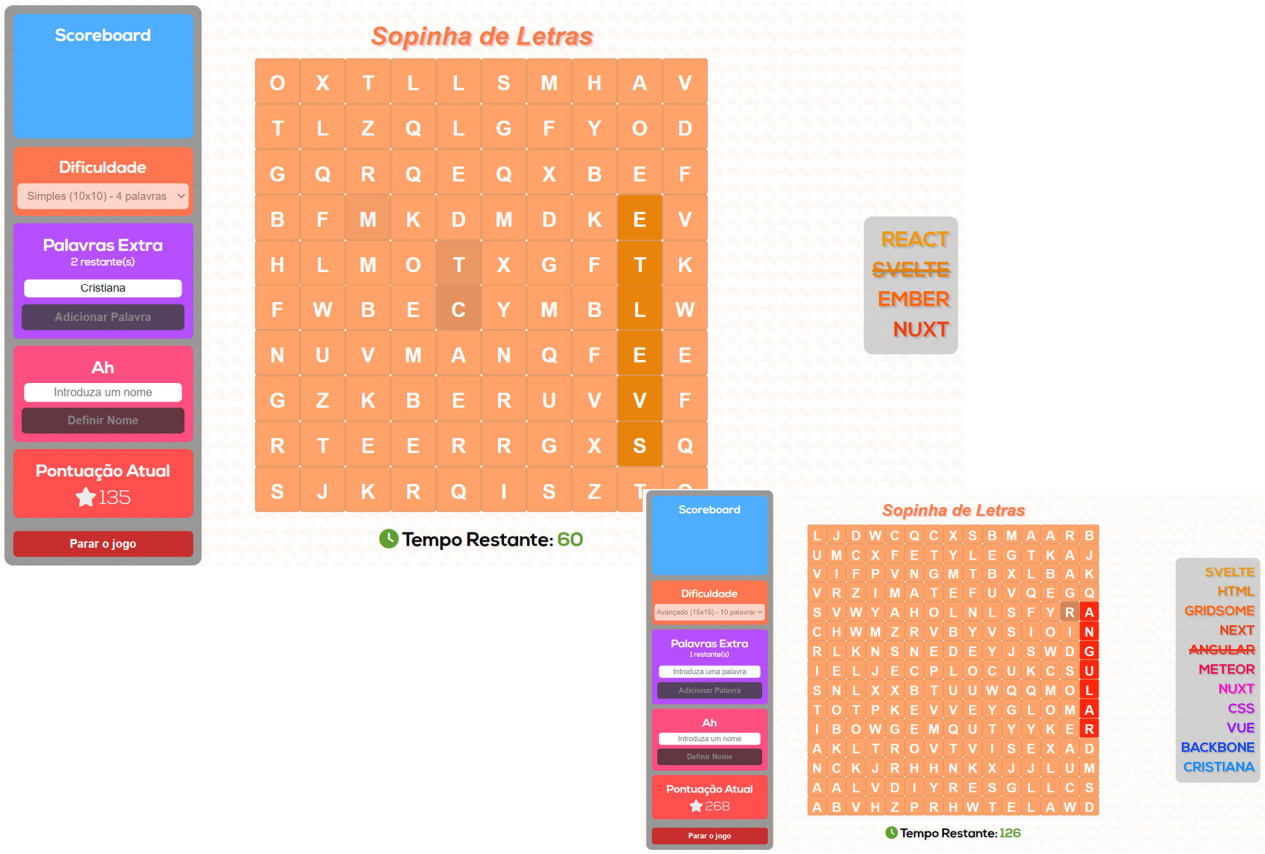




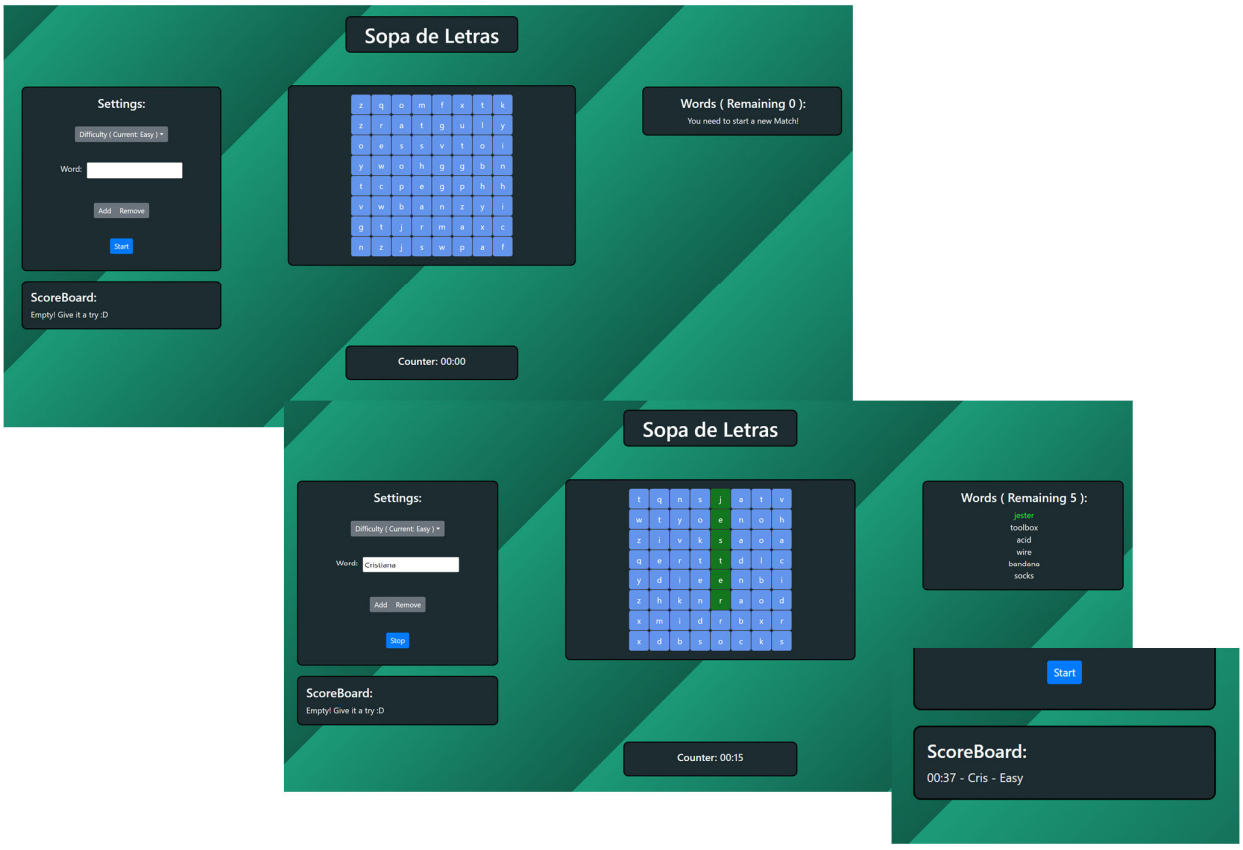




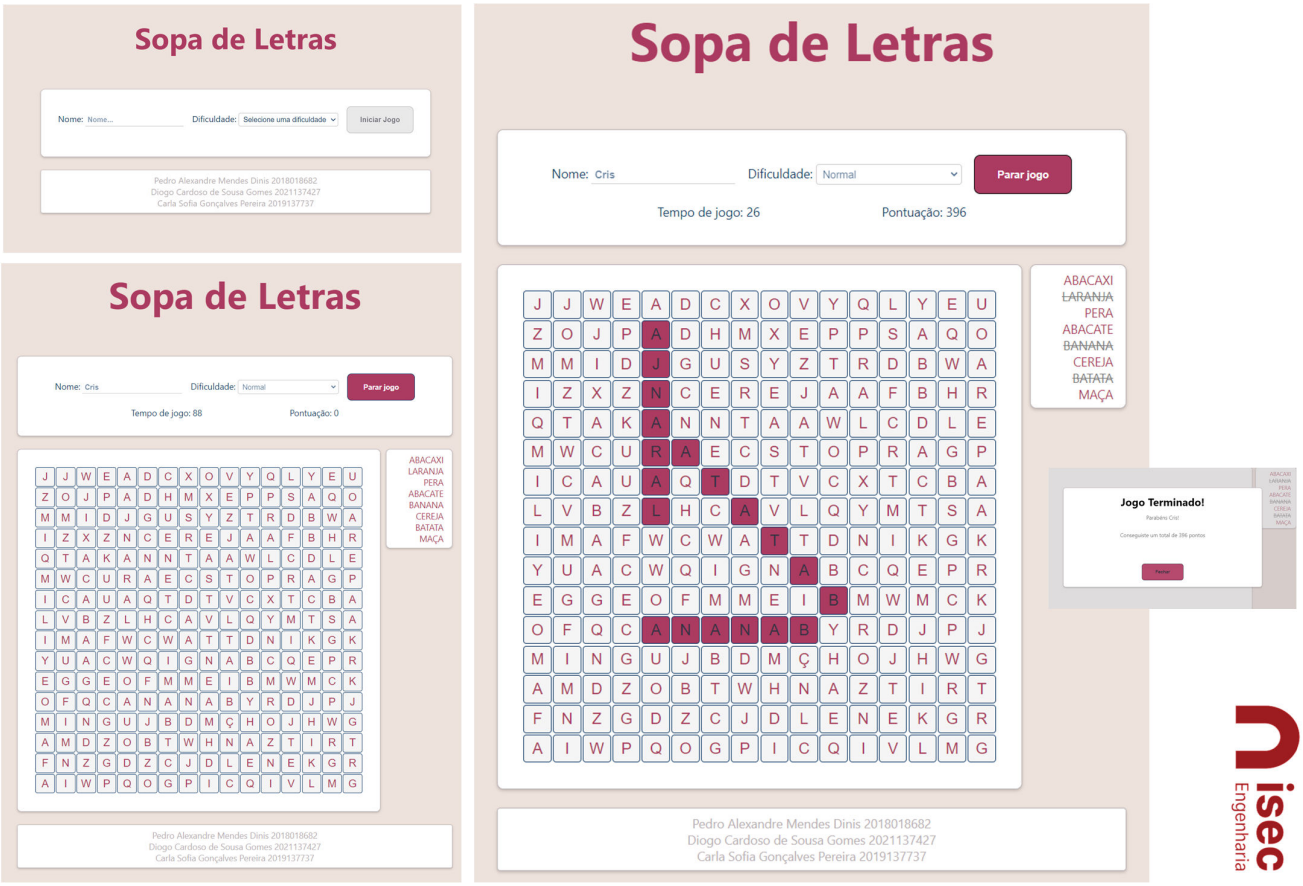
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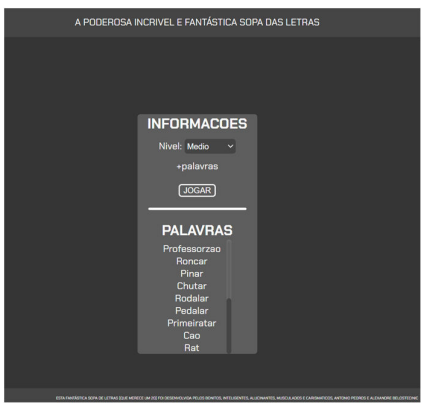
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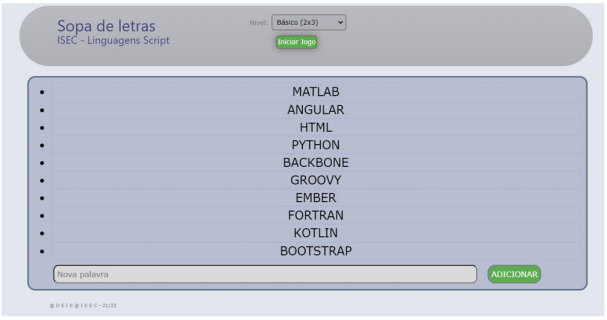
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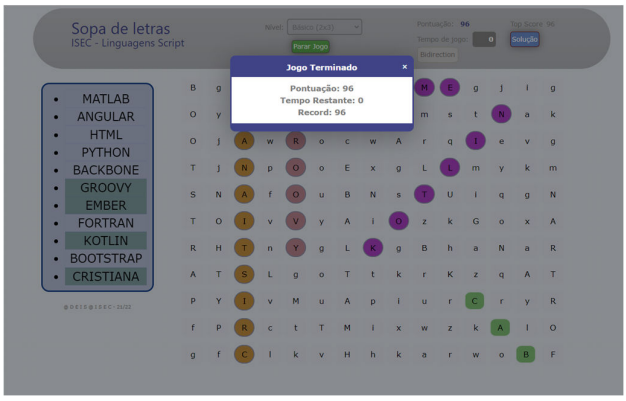
Trabalhos Práticos





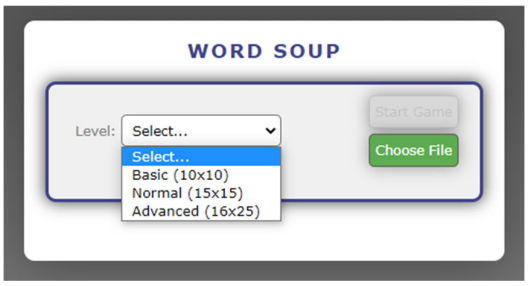


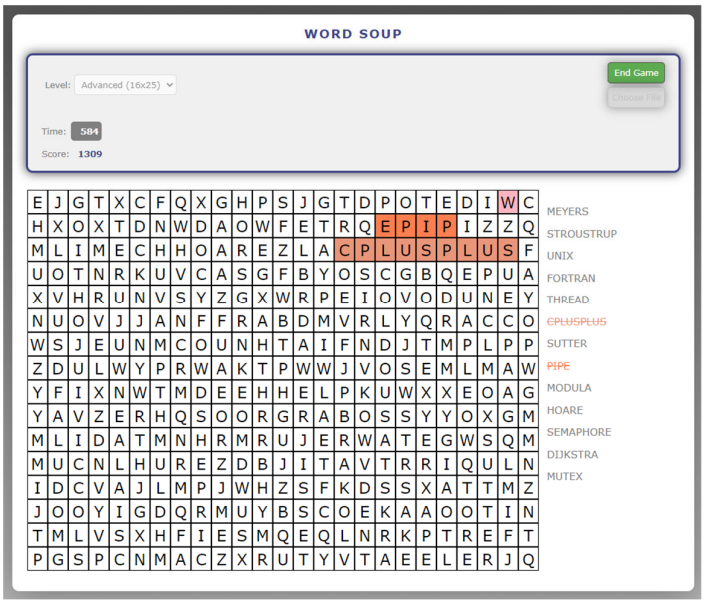


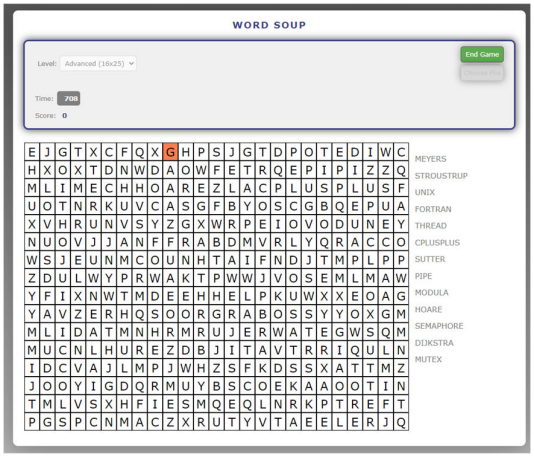


“Sopa de Letras”

Trabalhos Práticos







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